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A lightweight intrusion detection approach for CAN bus using depthwise separable convolutional Kolmogorov Arnold network

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Ensuring the cybersecurity of modern vehicles is paramount as connected and autonomous systems become increasingly prevalent. However, existing intrusion detection systems (IDS) often face challenges such as imbalanced datasets and high computational demands, limiting their practical deployment in automotive environments. To address these limitations, we employ spectral normalization GAN to synthesize anomalous data, achieving a balanced distribution across four attack categories and normal traffic. We further propose a lightweight classification model, named Depthwise Separable Convolutional Kolmogorov–Arnold network (DSC-KAN), which incorporates Kolmogorov–Arnold (K–A) theorem to enhance efficiency while maintaining high classification performance. Experimental results demonstrate that our approach outperforms existing methods in accuracy and computational efficiency, offering a robust and practical IDS solution. The proposed method has the potential to significantly improve vehicle network security, ensuring safer and more reliable deployment of connected and autonomous driving technologies.

Keywords Controller area network, In-vehicle security, Intrusion detection system, Generative adversarial nets, Kolmogorov–Arnold network

Over the past few decades, the rapid advancement of automotive electronics has revolutionized vehicle systems^{1–3}. One notable development is the widespread adoption of drive-by-wire (DBW) technology⁴, which replaces traditional mechanical linkages with electronic control systems to manage critical commands such as throttle, braking, and steering. Among the various in-vehicle systems, the Electronic Control Unit (ECU) serves as the central computational core to orchestrate control functions, while the Controller Area Network (CAN) establishes a robust communication bridge for facilitating efficient data exchange between interconnected components. This paper focuses on the study of the CAN bus, which offers higher real-time performance, scalability, and cost-effectiveness compared to Local Interconnect Network (LIN) and FlexRay, making it widely used in automotive electronic systems⁵.

However, CAN lacks advanced security mechanisms^{6,7}, such as authentication and encryption. The absence of these protective measures allows cyber-attackers to inject forged messages, potentially posing severe threats to the vehicle's control systems. For instance, an attacker could alter a vehicle's gear state by injecting malicious messages associated with specific CAN identifiers for transmission control. Such forged messages can be introduced through various attack vectors, including direct access to the On-Board Diagnostics II (OBD-II) port, as well as remote methods exploiting vulnerabilities in infotainment systems, telematics units, or wireless communication interfaces. Recent studies have identified numerous security vulnerabilities in modern vehicle systems, including electric window controls, warning lights, airbags, and tire pressure monitoring systems (TPMS). Furthermore, the growing interconnectivity of vehicles amplifies these risks.

In a high-profile example of CAN vulnerabilities, cybercriminals exploited a critical flaw (CVE-2023-29389) to steal Toyota vehicles⁸. Experts from EDAG Group and Canis Automotive Laboratories revealed that attackers leveraged CAN injection techniques to impersonate smart key receivers⁹, sending malicious commands to the vehicle's security systems. This allowed them to bypass anti-theft mechanisms, unlock the car, and disable the engine immobilizer. The incident underscores the urgent need for robust authentication protocols within

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CAN-based systems to thwart such malicious activities. In 2015, cybersecurity researchers Chris Valasek and Charlie Miller executed a groundbreaking attack on a Jeep Cherokee SUV¹⁰, exposing critical weaknesses in its CAN network. By accessing the system wirelessly, the attackers sent malicious commands to essential control units, including the engine, transmission, brakes, and steering. This attack culminated in the vehicle careening off the road into a ditch, vividly demonstrating the life-threatening implications of such vulnerabilities. The severity of the breach prompted Chrysler to recall over 1.4 million vehicles, marking the first instance of a vehicle recall driven by cybersecurity concerns. In 2016 and 2017, researchers from Keen Security Lab¹¹ demonstrated remote attacks on Tesla vehicles, revealing significant vulnerabilities in their systems. By exploiting a zero-day vulnerability in the Linux-based operating system of the Central Information Display (CID) module, attackers gained unauthorized access to the infotainment system via the cellular network. This breach enabled them to manipulate the central gateway, sending fraudulent CAN messages and UDS control signals to compromise critical vehicle functions. This demonstrates the potential for non-physical and network-based intrusions and the necessity of stringent software security in the connected automotive ecosystem.

Several defense mechanisms have been proposed to enhance the security of CAN bus systems against cyberattacks. Message authentication mechanisms¹², such as message authentication codes (MACs) or digital signatures, aim to ensure the integrity and authenticity of CAN communication. However, these methods often face challenges related to computational overhead and real-time constraints, especially in resource-constrained environments like embedded automotive systems. Encryption techniques are employed to safeguard the confidentiality of CAN messages but may suffer from similar performance issues and are often incompatible with the bandwidth limitations of the CAN protocol^{13–15}. Rule-based anomaly detection method¹⁶ relies on predefined patterns of normal behavior to identify attacks. However, these systems are relatively inflexible and prone to false positives when facing dynamic or complex cyber-attack patterns. In addition to software solutions, hardware-based protections have been proposed to protect CAN networks. These include security gateways or hardware security modules (HSMs)^{17–20} that physically isolate critical electronic control units (ECUs) and filter unauthorized traffic. Although hardware solutions can effectively mitigate certain attacks, they are difficult to update and upgrade, making it challenging to rapidly adapt to new attack manners.

Given these limitations, we propose a deep learning-based intrusion detection system (IDS) as a flexible and robust alternative. A key feature of our method is the transformation of raw CAN bus data into a grid-like structure, enabling its treatment as spatial information. Then, the DSC-KAN is designed and employed as a deep learning-based IDS to strike a balance between detection accuracy and computational efficiency. By employing the DSC-KAN algorithm, our IDS can adapt to evolving threats and provide comprehensive protection for CAN bus systems. This paper presents the following contributions:

- To the best of our knowledge, this work is the first study employing the KAN to IDS for the in-vehicle network security;
- SN-GAN is employed for data augmentation to handle the class imbalance between normal and abnormal data;
- A novel DSC-KAN is designed inspired by depthwise separable convolutions to strike a commendable balance between model complexity design and detection accuracy.

Literature review

In this section, we comprehensively review recent advancements in vehicle network attack detection, with particular emphasis on intrusion detection algorithms, data augmentation techniques, and cutting-edge developments in deep learning.

IDS

In modern vehicles, the CAN protocol serves as a widely adopted communication standard, facilitating data exchange among various electronic control units (ECUs). However, with the increasing complexity of automotive networks, cybersecurity concerns have become a pressing challenge. Rule-based IDSs were among the earliest approaches deployed to ensure vehicular network security. These methods typically rely on predefined rules or patterns to determine whether messages on the CAN bus adhere to specified standards, thereby detecting potential intrusions. For instance, Larson et al.²¹ leveraged the object dictionary outlined in the CANopen standard draft 3.01 to establish security specifications for communication and ECU behavior, exploring the applicability of rule-based methods in detecting cyberattacks within in-vehicle networks. Similarly, Arman et al.²² utilized fuzzy association rules to construct an attack detection classifier. Specifically, they assigned the sample's label based on the best-matching rule set by evaluating the compatibility of incoming samples with rule sets corresponding to different categories using specific matching metrics. Some other researchers identify different attack patterns by analyzing the statistical characteristics of malicious cyber-attacks. For example, Khan et al.²³ employed threshold computation and statistical characteristics of attacks to exploit malicious patterns. Meanwhile, Muter and Asaj²⁴ proposed an entropy-based anomaly detection approach, defining entropy metrics on the CAN bus and comparing them against reference sets to identify attacks. Despite their contributions, rule-based methods face the following limitations due to the evolving complexity of attack strategies and the dynamic nature of vehicular network environments: (1) Reliance on predefined rules or thresholds: Sophisticated attackers can manipulate message formats or frequencies to mimic normal communication, thereby bypassing rule-based detection mechanisms; and (2) Focus on simple data features: Attributes such as message frequency or length are often prioritized, neglecting complex nonlinear relationships within the CAN data. As a result, these methods struggle to identify advanced attack patterns, such as coordinated or multi-stage attacks.

To address these challenges, an increasing number of studies shifted toward machine learning (ML) and deep learning (DL) approaches to enhance the detection accuracy of IDSs. Alfaridus et al.²⁵ employed several

novel ML-based methods, such as backpropagation neural networks and support vector machines, and achieved a detection accuracy of 99% for fuzzy attacks on the CAN bus. Hossain et al.²⁶ introduced a convolutional neural network (CNN)-based IDS to identify malicious network traffic in CAN systems. Casillo et al.²⁷ employed Bayesian networks to rapidly detect malicious messages on the CAN bus by incorporating contextual and situational awareness. More recently, Song et al.²⁸ developed an IDS based on deep convolutional neural networks (DCNNs) to learn traffic patterns and detect malicious flows without relying on manual feature engineering. Their DCNN model was optimized to simplify the Inception-ResNet architecture by eliminating unnecessary neurons. Nonetheless, existing IDSs often simplify the anomaly detection to a binary classification problem, which only indicates whether a cyber-attack has occurred, without identifying the specific type of attack. Furthermore, in real-world scenarios, attacked data in CAN is extremely scarce, with the majority of data being benign. This severe imbalance between positive and negative labels presents a critical challenge. Finally, compared to attack-specific IDSs, a generalized IDS capable of detecting various attack types in real time offers greater practical utility.

Data augmentation

Generally, the performance of CAN IDS heavily relies on the availability of high-quality training data. However, the scarcity of attacked data in real-world scenarios remains a major bottleneck in developing an efficient IDS model. To address this challenge, Data Augmentation (DA) can be regarded as a promising alternative to expand existing datasets by generating, transforming, or synthesizing new data samples, thereby enhancing the model's generalization ability and robustness.^{29–31} Traditional data augmentation methods primarily rely on simple mathematical transformations and interpolation techniques to generate new samples^{32,33}. These approaches include random perturbations and oversampling techniques³⁴. Random perturbation introduces noise into the original data, such as Gaussian noise or salt-and-pepper noise, creating slightly altered samples. This method is easy to implement and is suitable for numerical data; however, it can introduce uncontrollable noise, which may cause the generated data to deviate from the true distribution. Another common traditional method is oversampling, with the most well-known being the Synthetic Minority Over-sampling Technique (SMOTE)³⁵. SMOTE generates new samples to balance the class distribution by interpolating between minority class samples. However, this method can lead to overfitting high-dimensional data and may generate invalid samples outside the true data distribution boundary. The Adaptive Synthetic Sampling (ADASYN) method³⁶, an improved version of SMOTE, adaptively generates data based on the complexity of the sample distribution. However, its preference for difficult samples may result in unstable quality of the newly generated data.

Data augmentation driven by deep learning can extract features from high-dimensional complex data to generate realistic samples. As a commonly used method, Autoencoder (AE)³⁷ maps input data into a lower-dimensional latent space, from which data samples are reconstructed. However, due to the inherent limitations in the design of the encoder and decoder, the diversity of the generated samples tends to be relatively low. Variational Autoencoder (VAE)^{38,39} extends this approach by introducing probabilistic assumptions on the latent space, enabling the sampling of new data points from the latent space. Compared to AE, VAE-generated samples exhibit greater diversity, but the model's ability to capture complex distributions is also limited. Time-series generative models^{40,41}, such as those based on Recurrent Neural Networks (RNNs) or Transformers, have been specifically designed to augment imbalanced datasets. These models capture temporal dependencies within the dataset to generate samples that align with the underlying distribution. However, these methods always involve complex model structures, which may hinder their generalization capabilities on smaller datasets. As a promising alternative, Generative Adversarial Network (GAN)⁴² consists of a generator and a discriminator in an adversarial setup, capable of generating highly realistic data, particularly images, text, and audio. Specifically, the generator produces synthetic data from a latent space, while the discriminator learns to distinguish between real and synthetic data, collectively improving the quality and authenticity of the generated samples.

Deep learning approaches

In recent years, deep learning technologies have been evolving at an astonishing pace^{43–45}. Convolutional Neural Network (CNN), introduced by LeCun et al.⁴⁶, has become widely adopted due to their ability to handle high-dimensional image data. The Vision Transformer (ViT)⁴⁷ leverages the transformer architecture for image classification by treating image patches as sequences, similar to how text is processed in natural language tasks. In 2024, a network architecture distinct from the traditional Multilayer Perceptron (MLP) was introduced, that is, the Kolmogorov-Arnold Network (KAN)⁴⁸. Unlike conventional deep learning paradigms, which first apply linear transformations followed by nonlinear activation functions, KAN reverses this order by performing nonlinear activation on the features before applying linear weighting. Notably, the activation functions in KAN are learnable parameters. Building on this framework, Bodner⁴⁹ proposed the Convolutional KAN, a novel network layer that integrates the learnable nonlinear activation functions from KANs into convolutional operation. Comparative performance evaluations on the Fashion-MNIST dataset demonstrated that the convolutional KAN achieves similar accuracy while utilizing only half the number of parameters compared to traditional CNNs. The KAN convolution kernels are capable of learning more detailed information, opening new avenues for network attack detection in vehicular systems.

Furthermore, CAN bus anomaly detection models operate in resource-constrained environments. Therefore, we analyze the necessity of lightweight networks from the following aspects: (1) Computational resource limitations: On-board devices typically face limitations in computation resource, especially in embedded systems and low-power devices, which cannot support the computational demands for large-scale deep learning models; (2) Real-time requirements: For autonomous driving and safety-critical scenarios, anomaly detection must possess rapid response capabilities. Traditional large-scale deep learning models suffer from response delays and compromise system real-time performance. (3) Energy consumption: The substantial computational

load of large deep learning models typically requires higher power consumption, which presents a significant challenge for vehicular systems, especially in electric vehicles. Therefore, besides considering the accuracy of deep learning methods, this study also focuses on designing a lightweight IDS.

Method

Spectral normalization GAN (SN-GAN)

In the real world, the majority of the CAN bus dataset represents normal operating conditions, while malicious attack data is extremely scarce, leading to a class imbalance problem. In this condition, directly applying deep learning models for classification may result in overfitting to normal data, significantly reducing the model's ability to detect attack samples. To address this issue, this study innovatively employs GANs for data augmentation to effectively mitigate the data imbalance problem. Specifically, GANs can generate realistic malicious attack samples, thereby balancing the number of positive and negative samples and enhancing the model's ability to identify attack behaviors. By generating diverse attack samples, GANs can simulate various attack scenarios, improving the model's generalization capability and robustness.

GANs are trained through a competitive process to generate samples that resemble the true data distribution through adversarial interaction between a generator and a discriminator. The generator aims to create fake samples that closely resemble real data, thus achieving the effect of "deceptive realism". The discriminator, on the other hand, tries to determine whether the input sample originates from the real dataset or from the generator. The training process of a GAN can be viewed as a zero-sum game, with the generator and discriminator assuming roles akin to an attacker and a defender, respectively. The generator's objective is to deceive the discriminator as effectively as possible, making it unable to distinguish the generated samples, while the discriminator works to improve its discriminatory ability. This adversarial game continues until the generator produces sufficiently realistic samples such that the discriminator cannot accurately distinguish between the real and generated data, reaching a state of what is known as Nash equilibrium. At this point, the generator and discriminator have reached a relative balance, and the generated samples are nearly indistinguishable from real data, marking the successful training of GANs.

However, during the training of GANs, the discriminator's gradient is often unstable, particularly in high-dimensional spaces. As a result, vanishing and exploding gradients frequently arise, leading to the phenomenon of mode collapse, where the generator and discriminator fail to train effectively. Ultimately, the trained generator may produce samples with limited diversity or samples that bear little resemblance to real data. Fortunately, Miyato et al.⁵⁰ introduced the Spectral Normalization GAN (SN-GAN) with a novel weight normalization technique (i.e., spectral normalization) to stabilize the training of the discriminator. Spectral normalization controls the Lipschitz constant of the discriminator by restricting the spectral norm of each of its layers. This approach is computationally simple and can be seamlessly integrated into existing GAN.

Lipschitz continuity is a concept used to describe the rate at which a function changes. A function $f(\cdot)$ is said to be K -Lipschitz continuous if there exists a constant K such that for all x_1 and x_2 , the following inequality holds:

$$\|f(x_1) - f(x_2)\| \leq K \|x_1 - x_2\| \quad (1)$$

In the context of GANs, we aim for the discriminator D to be 1-Lipschitz continuous, meaning:

$$\|D(x_1) - D(x_2)\| \leq \|x_1 - x_2\| \quad (2)$$

The formulation (2) ensures that the discriminator is not overly sensitive to minor variations in the input, thus enhancing the training stability. In Wasserstein GANs (WGAN)⁵¹, a similar approach involves clipping the weights of the discriminator within a constrained range (e.g., $[-0.01, 0.01]$) after each training step. A variant of WGAN, known as WGAN-GP⁵², replaces weight clipping with gradient penalty to enforce the Lipschitz constraint. WGAN-GP adds a gradient penalty to the loss function, ensuring smooth and stable gradients. In contrast to explicitly regularizing the spectral norm by adding a regularization term to the objective function, Spectral Normalization (SN) directly normalizes the discriminator's parameters using the spectral norm to guarantee 1-Lipschitz continuity. Specifically, this is achieved through the following equation:

$$\overline{W}_{SN}(W) = \frac{W}{\sigma(W)} \quad (3)$$

where W is the weight matrix of the discriminator, and $\sigma(W) = \max_{h: \|h\|_2=1} \|Wh\|_2$ is the largest singular value of the matrix W . By dividing the weight matrix by its spectral norm, the normalized weight matrix $\overline{W}_{SN}(W)$ ensures that the 1-Lipschitz continuity condition is satisfied, i.e., $\|\overline{W}_{SN}(W)\|_{Lip} \leq 1$. In practical applications, computing the spectral norm involves performing a singular value decomposition (SVD) on the matrix, which typically requires $O(n^3)$ operations, where n is the matrix dimension. This computation becomes prohibitively expensive, especially when dealing with the large parameter matrices often encountered in deep learning models. Thus, the power iteration method is served as an approximation technique for calculating the spectral norm. The basic steps of the power iteration method are shown in Algorithm 1.

Input: the discriminator's weight parameter W , the minimum threshold $\Delta_{\min}$

Output: the approximate spectral norm for W

1: Select a random vector $v_0 \in R^n$, set $Flag1=1, k=0$

2. **While** $Flag1==1$ **do**

3. $k = k + 1$

4. $v_k = Wv_{k-1}$

5. **If** $\|v_k - v_{k-1}\| \leq \Delta_{\min}$ **then**

6. set $Flag1=0$

7. **End If**

8. **End While**

9. $\lambda = v_k^T W v_k$

10. **Return** λ

Algorithm 1. Fast approximation of the spectral norm.

The power iteration method offers an efficient approach to approximating the largest singular value, reducing the computational complexity from $O(n^3)$ to $O(kn)$, where k is the iteration number. Notably, spectral normalization is typically not applied to the generator, as its training process is relatively stable compared to that of the discriminator. Both SN-GAN and the standard GAN aim to maximize the generator's adversarial loss while minimizing the discriminator's adversarial loss. The specific loss functions are as follows:

$$\min_G \max_D \mathbb{E}_{x \in p_{data}(x)} [\log D(x)] + \mathbb{E}_{x \in p_g(x)} [1 - \log D(x)] \quad (4)$$

According to formulation (4), the discriminator D aims to maximize the loss function (i.e., correctly distinguishing between real and generated data), while the generator G seeks to minimize the same loss function (i.e., generating data that the discriminator incorrectly classifies as real). In theory, the generator and discriminator will eventually reach a Nash equilibrium, where the generated data cannot be distinguished from real data by the discriminator, and the discriminator can no longer improve its classification accuracy. At this point, the distribution of generated data $p_g(x)$ will closely approximate the distribution of the real data $p_{data}(x)$. In this work, we leverage SN-GAN to generate a large amount of CAN bus data injected with cyber-attacks. The detailed SN-GAN is shown in Fig. 1.

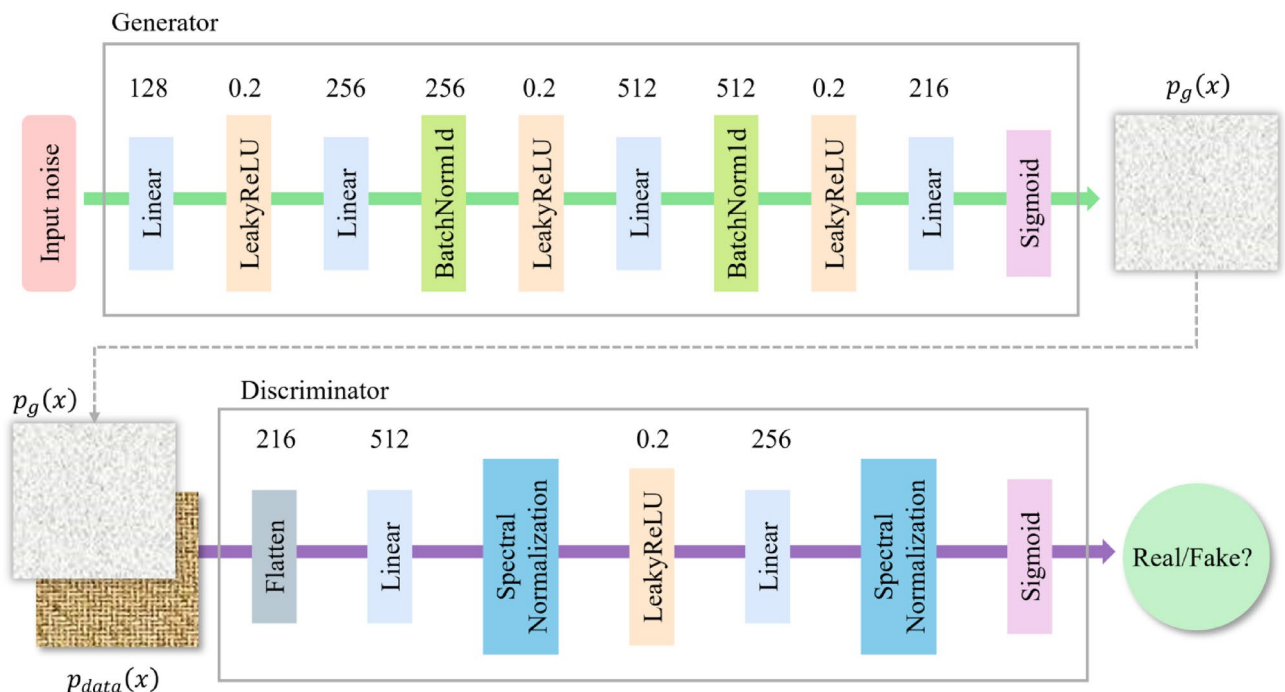


Fig. 1. The architecture of SN-GAN for generating attacked CAN bus data.

Depthwise separable convolutional KAN

Liu et al.⁴⁸ proposed a novel KAN network in 2024 and attracted much attention in the deep learning field. Unlike MLPs, KANs are grounded in the Kolmogorov-Arnold (K-A) theorem, utilizing non-linear function decomposition for data fitting. The key insight from the K-A theorem is that any continuous multivariate function $f(x_1, x_2, x_3, \dots, x_n)$ can be decomposed into a set of continuous univariate functions. Specifically, there exists a set of continuous univariate functions $\varphi_i(\cdot)$ and $\varphi_{ij}(\cdot)$ that allow the multivariate function to be expressed as a linear combination of these univariate components, as shown in Eq. (5).

$$f(x_1, x_2, \dots, x_n) = \sum_{i=1}^{2n+1} \phi_i \left(\sum_{j=1}^n \phi_{ij}(x_j) \right) \quad (5)$$

In theory, this decomposition enables efficient modeling of complex relationships by breaking down high-dimensional functions into one-dimensional (1D) forms. However, these 1D functions exhibit non-smooth or even fractal behaviors, making them difficult to learn in practice. Fortunately, Liu et al.⁴⁸ address this problem by extending the original K-A theorem which restricts KANs to a simple two-layer network with an input dimension n and a middle layer width of $2n + 1$. They proposed a more flexible approach, allowing KAN to become both wider and deeper. Consider a specific KAN represented by the sequence $\{n_0, n_1, \dots, n_l, \dots, n_L\}$, where n_l denotes the number of neurons in the l -th KAN layer and $x_{l,i}$ denotes the activation value of the i -th neuron in the l -th layer. Between layer l and layer $l + 1$, there exist $n_l \times n_{l+1}$ distinct activation functions: $\varphi_{l,i,j}(\cdot)$, $i = 1, 2, \dots, n_l$, $j = 1, 2, \dots, n_{l+1}$. The pre-activation of $\varphi_{l,i,j}(\cdot)$ corresponds directly to $x_{l,i}$, and the activation value of j -th neuron in $l + 1$ -th layer is then computed as the sum of all post-activation values from the previous layer. In matrix form, this can be expressed as:

$$\mathbf{x}_{l+1} = \underbrace{\begin{pmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \cdots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \cdots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{l,n_{l+1},1}(\cdot) & \phi_{l,n_{l+1},2}(\cdot) & \cdots & \phi_{l,n_{l+1},n_l}(\cdot) \end{pmatrix}}_{\Phi_l} \mathbf{x}_l \quad (6)$$

Inspired by KAN, the Convolutional KAN (CKAN) has been proposed, where traditional convolutional layers are replaced by KAN convolutional layers. The primary difference between KAN convolutional layers and traditional convolutional layers lies in the convolution kernels. In the latter, the kernels consist of weight parameters, whereas in the former, each kernel element is replaced by a learnable nonlinear function based on B-splines. Typically, a two-dimensional image is represented as a matrix IM , where each element $a_{k,l}$ represents the pixel value at position (k, l) in the image:

$$IM = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,p} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,p} \end{pmatrix} \quad (7)$$

In convolution operations, a convolutional kernel or filter is slid over the entire image, computing the weighted sum of the local region of the image at each position, thereby generating a new output feature map. Unlike traditional convolutional methods, KAN convolution applies a set of nonlinear functions to the corresponding pixels in the input image, followed by a summation. Mathematically, this can be expressed as:

$$(IM * K)_{i,j} = \sum_{k=1}^N \sum_{l=1}^M \phi_{k,l}(a_{i+k,j+l}) \quad (8)$$

where N and M represent the height and width of the kernel K , and $\varphi_{k,l}$ denotes the learnable nonlinear function of K . This approach allows KAN convolution to capture complex nonlinear relationships within image data, thereby enhancing model performance. However, the large number of learnable parameters associated with the B-spline functions in the convolutional KAN introduces significant computational overhead, limiting its applicability in real-time embedded vehicle IDS systems. To address this challenge, we propose the Depthwise Separable Convolutional KAN (DSC-KAN), which decomposes the KAN convolutional layer into two distinct processes (illustrated in Fig. 2): the depthwise KAN convolution and the pointwise KAN convolution. In the depthwise KAN convolution, each kernel operates on a single channel, ensuring that the output feature map retains the same number of channels as the input feature map. The pointwise KAN convolution, analogous to conventional convolution, employs a 1×1 kernel, with the number of channels corresponding to the output of the depthwise KAN convolution. Intuitively, the depthwise KAN convolution extracts feature along the spatial dimension, while the pointwise KAN convolution aggregates features along the channel dimension. By decomposing the standard convolution into depthwise and pointwise KAN convolutions, the DSC-KAN significantly reduces both computational cost and parameter count, while maintaining model performance and improving computational efficiency.

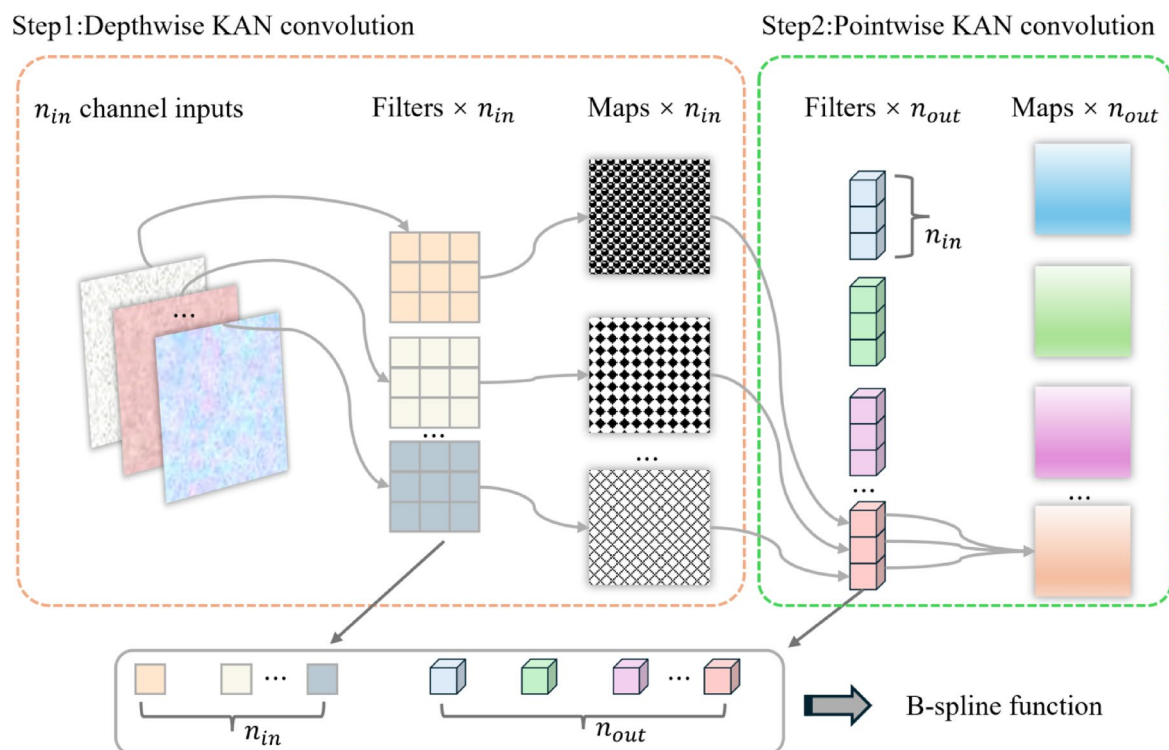


Fig. 2. The illustration of DSC-KAN.

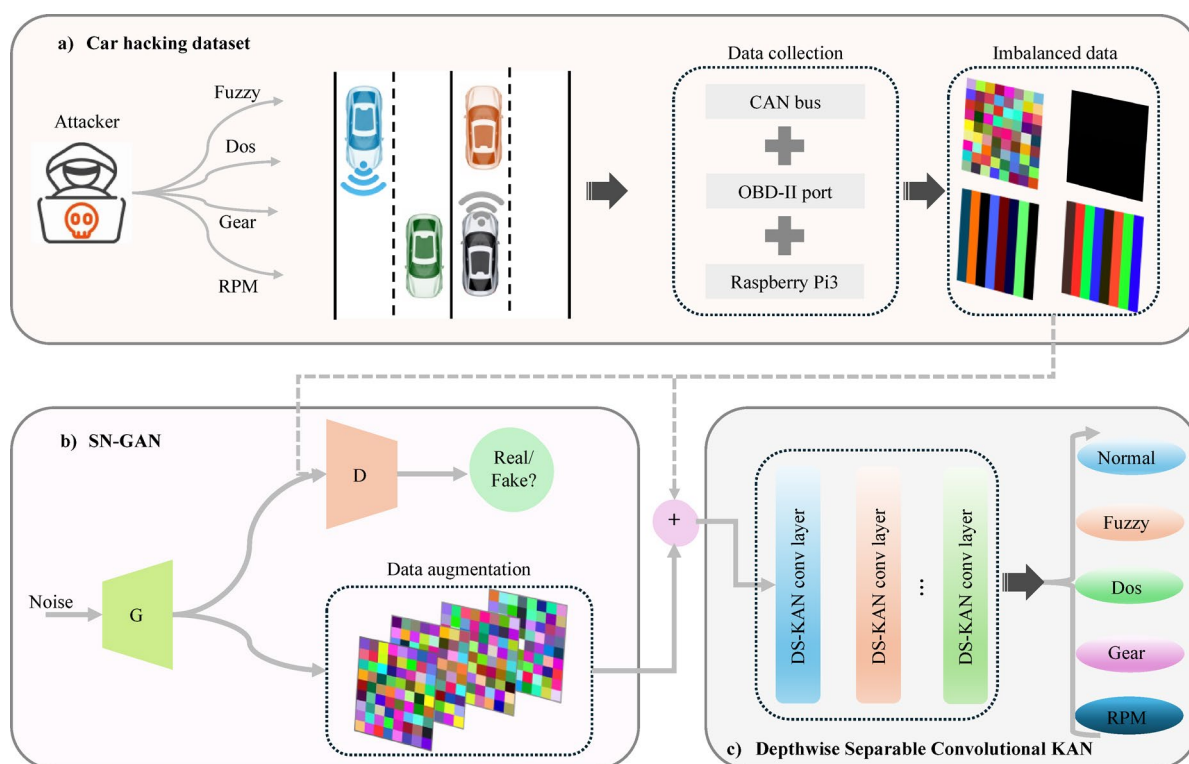


Fig. 3. The overall IDS for CAN bus.

Integrated IDS

To mitigate the threat posed by cyberattacks that could manipulate vehicle systems, this study integrates the above techniques for constructing an IDS for CAN bus, as illustrated in Fig. 3. Firstly, we consider four prevalent and highly damaging attack types: fuzzy attacks, DoS attacks, gear attacks, and RPM attacks. Fuzzy attacks involve sending malformed or anomalous data onto the vehicle’s CAN bus, inducing unpredictable behavior in the vehicle control system, which could lead to system crashes or loss of control. DoS attacks, on the other hand, overwhelm the network bandwidth with a large volume of false data packets or requests, thereby blocking communication between the vehicle and external systems and disrupting the normal operation of the onboard system. Gear and RPM attacks specifically target the engine control systems by manipulating sensor data (such as gear positions or engine speeds) which could cause uncontrollable acceleration, deceleration, or abnormal driving. In this study, a combination of the CAN bus, OBD-II ports, and Raspberry Pi 3 is employed to collect so-called Car hacking dataset. The OBD-II port provides an interface to the vehicle’s diagnostic system, offering access to real-time performance data, fault codes, and sensor outputs. The Raspberry Pi 3, as a low-cost and flexible computational platform, is connected to a CAN bus adapter and OBD-II interface, allowing efficient data collection, storage, and analysis of attack data. Secondly, we employ SN-GAN for data augmentation of the Car hacking dataset to address the issue of class imbalance in the dataset. Lastly, we train our proposed DSC-KAN to build an IDS that balances both accuracy and computational efficiency.

Numerical experiments
Dataset description

In the experiment, the effectiveness of the proposed method is validated using the well-known CAN-Intrusion dataset, which is publicly available at <https://ocslab.hksecurity.net/Datasets/CAN-intrusion-dataset>. To capture the CAN bus communication, a Y-cable is inserted into the OBD-II port, which is located beneath the steering wheel of the modern YF Sonata. A Raspberry Pi 3 is then connected to the CAN bus, and a laptop communicates with the Raspberry Pi 3 via WiFi. The dataset consists of four types of network attacks: DoS, fuzzy, RPM, and gear attacks. In the case of DoS attacks, high-priority CAN messages (e.g., ‘0×000’ CAN ID data packets) are injected within a short time frame. Fuzzy attacks involve injecting forged messages with random CAN IDs and data values. RPM/gear attacks focus on injecting messages with specific CAN IDs related to RPM/gear information. After collecting approximately 10 min of driving data, both attacked and non-attacked traffic are captured and labeled.

This study generates a multi-channel matrix form of data frames by transforming the time-series patterns of CAN messages into spatial local correlations. Specifically, we extract the 8-bit identifier from each of the most recent 27 CAN messages and reshape them into a 3D matrix of size 9×8×3, where 9×8 represents the spatial layout of 72 identifiers. By doing so, the temporal identifiers on the CAN bus can be directly fed as input into the convolutional network. The details of the training and testing sets are presented in Table 1. In the training set, our anomaly dataset contains only 500 samples per attack type, while the normal samples amount to 5,000. This imbalance stems from the fact that most real-world data belongs to normal instances, which are more readily available. In the testing set, we include 5,000 samples for each of the four attack types and the normal scenario to thoroughly evaluate the generalization performance of the deep learning-based IDS.

Data augmentation by SN-GAN

In this section, we employ both SN-GAN and standard GAN to augment the training dataset from “Dataset description” section. The experiments are conducted on a workstation equipped with a GTX 1650Ti GPU. The hyperparameters for training SN-GAN, standard GAN, WGAN and WGAN-GP are as follows: a learning rate of 0.0005, a batch size of 32, and 100 iterations. To vividly compare the stability of SN-GAN and standard GAN during the training process, we visualize the loss curves of generator and discriminator for SN-GAN and standard GAN. The discriminator loss is further divided into two components: the fake loss and the real loss. Additionally, to assess the quality of the generated images, we randomly sample and examine the images generated by the generator at various iterations. The results of these experiments are presented in Fig. 4.

As shown in Figs. 4a–c, the loss curves corresponding to the generator, the discriminator’s real loss, and the discriminator’s fake loss in the standard GAN, WGAN and WGAN-GP exhibit significant fluctuations, failing to converge to stable values. In conjunction with Fig. 4e, it is evident that the samples generated by the standard GAN exhibit remarkably similar fixed patterns during training iterations, indicating the occurrence of mode collapse. This is because the discriminator repeatedly encounters similar fake samples, resulting in unstable gradients during training. Consequently, the discriminator may fail to provide meaningful gradient signals to guide the generator’s optimization. In contrast, as depicted in Fig. 4d, the fake and real losses of the discriminator in SN-GAN stabilize after approximately 60 iterations and converge around 0.6. This equilibrium between the

Attack type	Training		Testing	
	CAN message	Attack image	CAN message	Attack image
Dos attack	108,000	500	1,080,000	5000
Fuzzy attack	108,000	500	1,080,000	5000
RPM attack	108,000	500	1,080,000	5000
Gear attack	1,080,00	500	1,080,000	5000

Table 1. Train and test dataset.

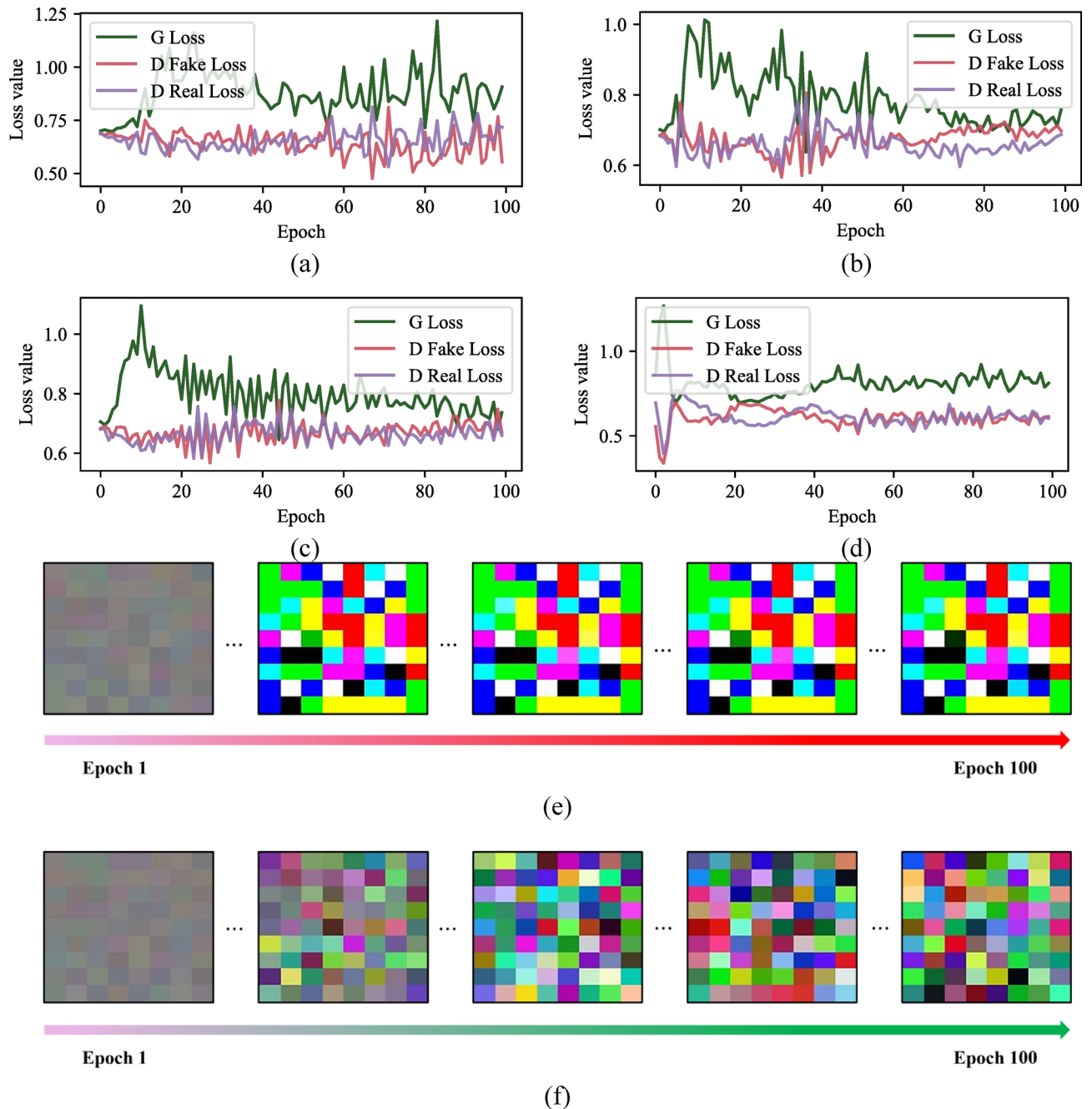


Fig. 4. Iterative loss curve for (a) standard GAN, (b) WGAN, (c) WGAN-GP and (d) SN-GAN along with examples of fuzzy attack data generated by (e) standard GAN and (f) SN-GAN.

generator and the discriminator signifies the attainment of a Nash equilibrium, where the discriminator can no longer easily distinguish between real and fake samples. Additionally, the quality of the attack data generated by the generator of SN-GAN progressively improves over the iterations, as shown in Fig. 4f. In summary, SN-GAN achieves greater training stability compared to standard GANs by leveraging spectral normalization.

As one of the most difficult-to-detect attacks, the fuzzy attack is selected to evaluate the efficacy of the anomaly data generated by our SN-GAN. In the CAN bus data frame, each frame consists of 8 bytes of data, with each byte ranging from 0 to 255. We train the SN-GAN on 500 instances of fuzzy attack data and subsequently generate 5000 additional samples using the well-trained generator. To assess the quality of the generated data, we compared the distribution histograms and probability density functions of the 5000 SN-GAN-generated samples against true samples. As shown in Fig. 5, the highly similar probability distributions indicate that the SN-GAN can generate a high-quality dataset that closely approximates the underlying distribution. This result validates the effectiveness of SN-GAN for attacked data augmentation.

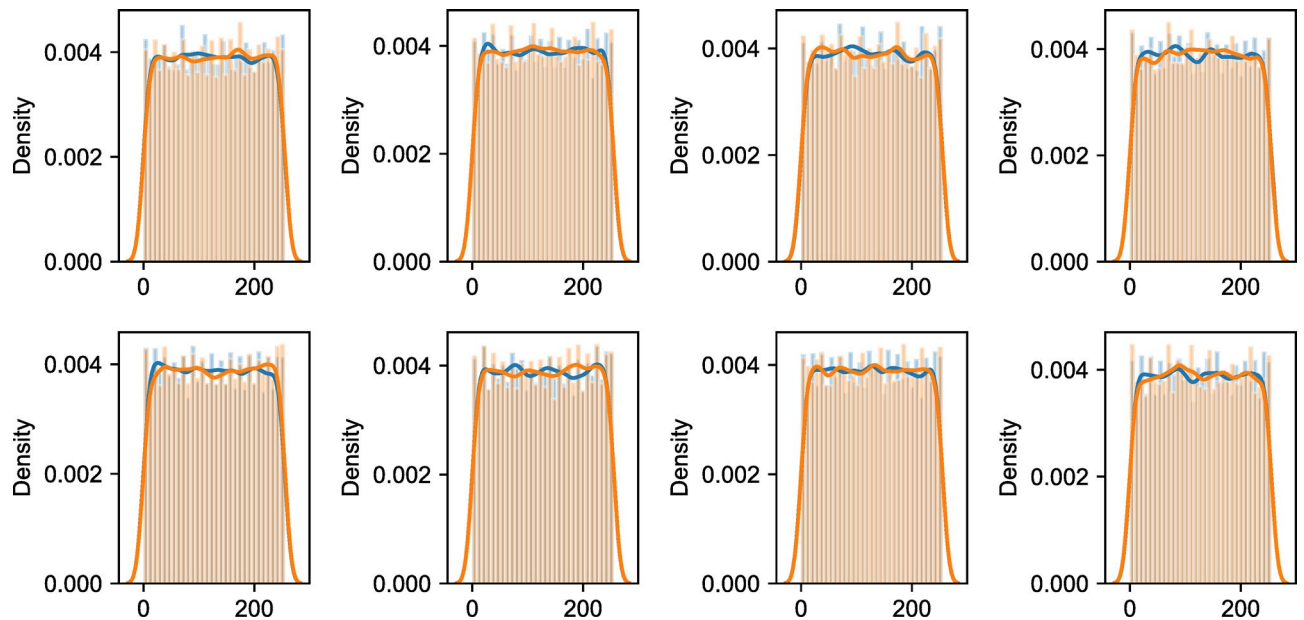


Fig. 5. Distribution histograms and probability density functions of the 5000 SN-GAN-generated samples on byte1 to byte8 against the truth samples.

Intrusion detection results

To develop an IDS that can be deployed in real-world scenarios, two key challenges must be addressed: (1) the scarcity of network attack events in real-world data and (2) the limited computational resources available on in-vehicle devices. For the former problem, we employ SN-GAN for data augmentation to expand the Car hacking dataset. For the latter problem, this section utilizes our proposed DSC-KAN to construct a lightweight IDS that balances both anomaly detection accuracy and computational efficiency. The performance of the IDS system is evaluated using three commonly used classification task metrics, that is, Precision, Recall, and F1-score. The above three performance indicators can be calculated as follows:

$$Precision = TP / (TP + FP) \quad (9)$$

$$Recall = TP / (TP + FN) \quad (10)$$

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall) \quad (11)$$

where TP (true positive) and TN (true negative) are the number of samples that are classified correctly as attack and normal, respectively. FP and FN are the number of samples that are classified incorrectly as attack and normal, respectively. Based on the above definitions, *Precision* measures the proportion of correctly predicted positive instances out of all instances predicted as positive. A high *Precision* value indicates fewer false positives, making it crucial in IDS where false alarms can directly affect the normal operation of the vehicle and reduce the trust of the driver or passengers. *Recall*, also known as sensitivity or true positive rate, measures the proportion of actual positive instances correctly identified by the model. A high *Recall* value implies fewer false negatives, which is especially critical in IDS because missed alarms may lead to further intrusion of the vehicle networking system by the attacker. In this condition, a missed alarm will cause more serious consequences than a false alarm. The *F1-score* is the harmonic means of precision and recall, offering a balanced measure that accounts for these two metrics.

In the following experiments, we incorporate the attack data generated by SN-GAN into the raw dataset and train the proposed DSC-KAN. The hyperparameters for training the DSC-KAN include a learning rate of 0.001, a batch size of 64, and 100 iterations. The degree of the B-spline employed in the activation function is set to 3. To evaluate the effectiveness of data augmentation using SN-GAN, we also train DSC-KAN on the original dataset without augmentation as a baseline for comparison. The training process and test results for both two cases are shown in Fig. 6. As shown in Fig. 6, when data augmentation is applied, DSC-KAN converges faster, reaching near-zero training error (i.e., 100% accuracy) on both the training and test sets after approximately 60 epochs. In contrast, without data augmentation, the convergence of DSC-KAN is slower, and the accuracy on the test set exhibits fluctuating increases rather than a steady improvement. This is because, in the absence of data augmentation, DSC-KAN may be overfit to the specific patterns in the training data, leading to instability in the model's performance on the test set.

Considering the imbalanced proportions across normal and attacked data, we set the number of true attacked samples to range from 100 to 700 in intervals of 100. Based on these different training datasets, several independent DSC-KANs with DA are trained to obtain their F1 scores. The results, given in Fig. 7, demonstrate that the more true attacked samples there are, the higher the model's accuracy becomes. This is because an

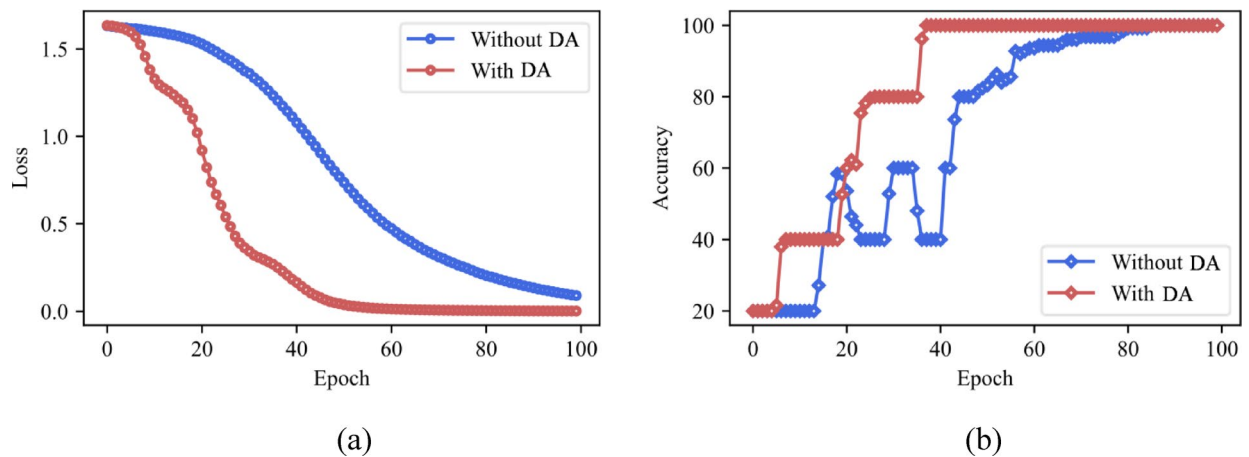


Fig. 6. The iterative loss curve (a) and accuracy curve (b) with and without data augmentation.

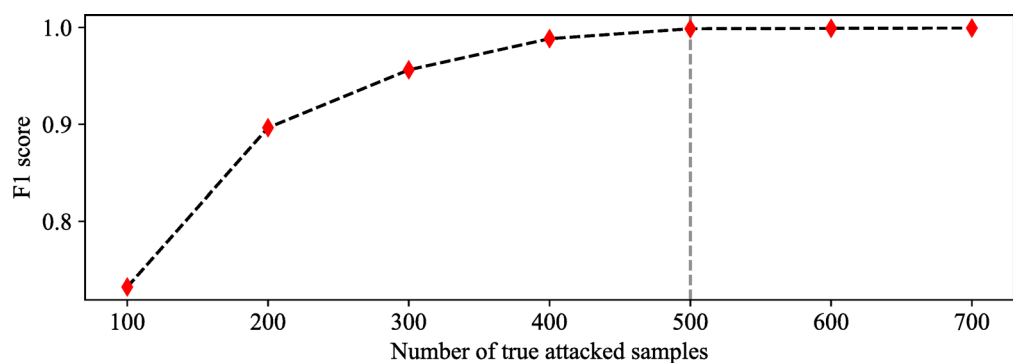


Fig. 7. F1 score of DSC-KAN with DA on training datasets with different number of true attacked samples.

	Precision	Recall	F1
Fuzzy			
DSC-KAN	1.00	0.82	0.90
DSC-KAN-DA	1.00000	0.99740	0.99870
Gear			
DSC-KAN	1.00	1.00	1.00
DSC-KAN-DA	0.99960	1.00000	0.99980
RPM			
DSC-KAN	0.88	1.00	0.94
DSC-KAN-DA	0.99761	1.00000	0.99880
DoS			
DSC-KAN	0.91	1.00	0.95
DSC-KAN-DA	1.00000	1.00000	1.00000
Normal			
DSC-KAN	0.95	0.90	0.92
DSC-KAN-DA	1.00000	0.99980	0.99990

Table 2. Precision, recall, and F1-score of DSC-KAN on the test set with and without data augmentation.

increased number of true attacked samples improves the training performance of SN-GAN, leading to the generation of higher-quality fake samples. Using the elbow method, a technique that identifies the point at which increasing complexity yields diminishing returns, we observe that the number of true attacked samples exceed 500 do not significantly enhance model performance. Therefore, this analysis supports the assumption of 500 true attacked samples as optimal for this dataset.

Table 2 presents the Precision, Recall, and F1-score of DSC-KAN on the test set with and without data augmentation. It is evident that without data augmentation, DSC-KAN achieves a recall of only 0.82 when detecting fuzzy attacks, resulting in a notably high false negative rate. This is primarily due to the nature of fuzzy attacks, which generate signals that are inherently uncertain and ambiguous. Consequently, in the absence of data augmentation, the limited training samples hinder DSC-KAN’s ability to recognize the diverse patterns of fuzzy attacks. Even more concerning is the fact that the DSC-KAN trained without data augmentation mistakenly classifies normal data as cyber-attacks. Such misclassifications can severely degrade the driver and passenger experience. For instance, the braking system might erroneously detect an attack, triggering an emergency brake without any actual threat. Similarly, the autonomous driving system could mistakenly stop the vehicle or alter its route without cause.

Finally, we compare the performance of the traditional CNN, Vision Transformer (ViT), C-KAN, and DSC-KAN models with data augmentation, as shown in Table 3; Fig. 8. It is evident that CNN and ViT exhibits lower recall and F1 scores compared to the two KAN-based models (C-KAN and DSC-KAN), highlighting the advantage of KAN convolutions in capturing complex nonlinear relationships within the Car hacking dataset. Moreover, although the model parameters of DSC-KAN are reduced from 2.8 M to 0.8 M compared to C-KAN, its anomaly detection performance only exhibits a slight decline. Notably, for the more challenging Fuzzy attacks, as shown in Fig. 8, C-KAN missed 6 out of 5000 test samples, while DSC-KAN missed 13, indicating only a minor performance degradation. These results demonstrate that the proposed DSC-KAN strikes a commendable balance between model lightweight design and detection accuracy.

Dimensionality reduction by T-SNE

In this experiment, we aim to evaluate the feature extraction capabilities of the trained DSC-KAN. To achieve this, we perform a forward pass through the network for each of the 5000 test samples, extracting their corresponding latent features from the final layer of the model. These features are then subjected to dimensionality reduction using t-Distributed Stochastic Neighbor Embedding (T-SNE) algorithm, reducing them to both 3-dimensional and 2-dimensional spaces for visualization purposes. The resulting visualizations in Fig. 9 reveal clear, discernible clustering of samples belonging to different attack types. Notably, in both the 2D and 3D representations, samples from each class formed tightly-knit clusters, which are easily distinguishable from those of other classes. This strong separation of feature representations in a lower-dimensional space indicates that DSC-KAN has effectively learned meaningful, discriminative features for distinguishing between five types of CAN data. We further compared the dimensionality-reduced data obtained from the fuzzy and Normal types and observed that the two datasets are closely adjacent and exhibit greater dispersion than the other three data types. This phenomenon can be attributed to the nature of the fuzzy attack, which employs fuzzification on the CAN

	Precision	Recall	F1
Fuzzy			
CNN-DA	1.00000	0.99080	0.99538
ViT-DA	1.00000	0.99320	0.99659
C-KAN-DA	1.00000	0.99880	0.99940
DSC-KAN-DA	1.00000	0.99740	0.99870
Gear			
CNN-DA	0.99960	1.00000	0.99980
ViT-DA	0.99780	1.00000	0.99890
C-KAN-DA	0.99980	1.00000	0.99990
DSC-KAN-DA	0.99960	1.00000	0.99980
RPM			
CNN-DA	0.99088	1.00000	0.99542
ViT-DA	0.99325	1.00000	0.99661
C-KAN-DA	0.99880	1.00000	0.99940
DSC-KAN-DA	0.99761	1.00000	0.99880
DoS			
CNN-DA	1.00000	1.00000	1.00000
ViT-DA	0.99980	1.00000	0.99990
C-KAN-DA	1.00000	1.00000	1.00000
DSC-KAN-DA	1.00000	1.00000	1.00000
Normal			
CNN-DA	1.00000	0.99960	0.99980
ViT-DA	1.00000	0.99760	0.99880
C-KAN-DA	1.00000	0.99980	0.99990
DSC-KAN-DA	1.00000	0.99980	0.99990

Table 3. Comparison results with different algorithms.

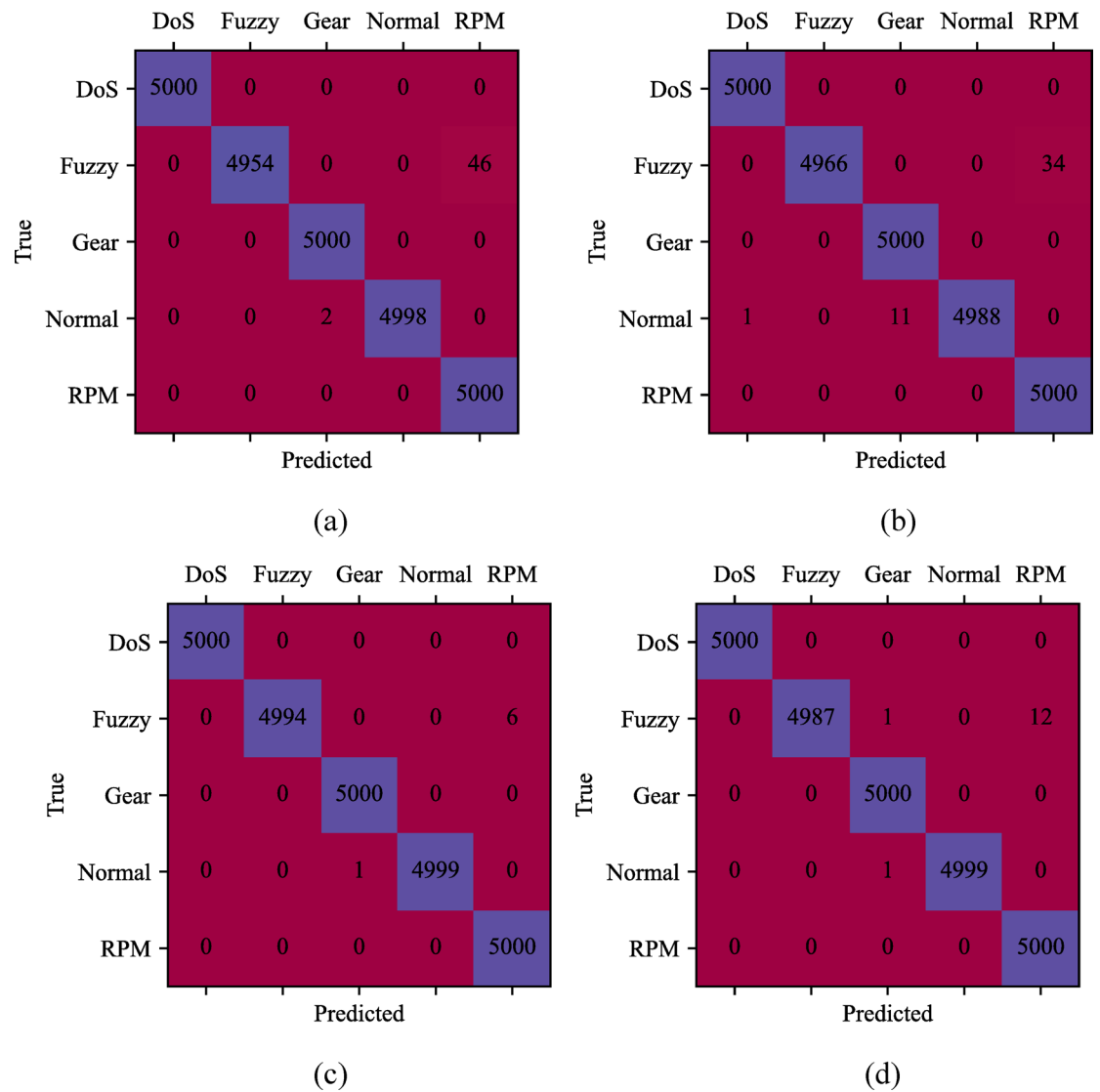


Fig. 8. Confusion matrix of (a) CNN, (b) ViT, (c) C-KAN and (d) DSC-KAN.

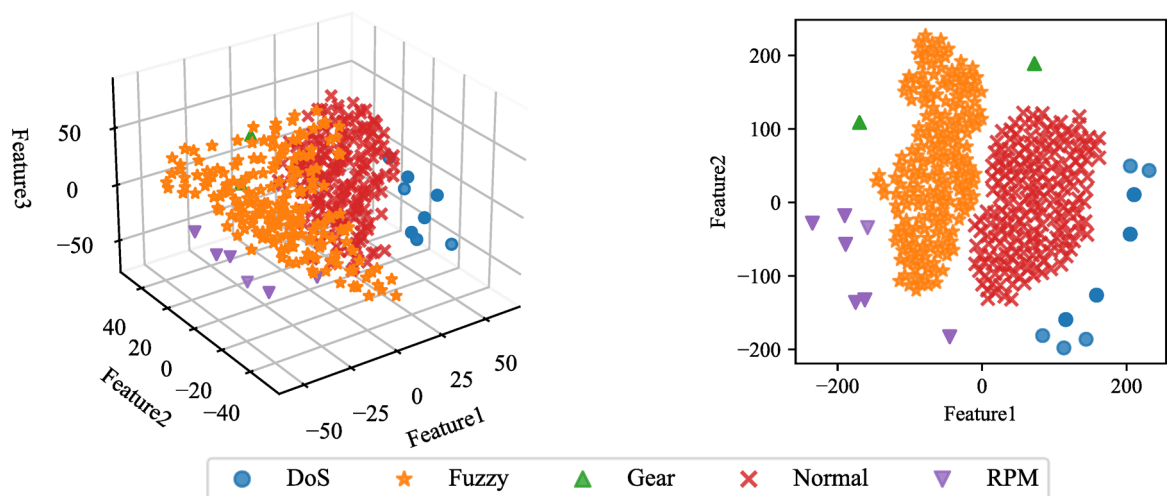


Fig. 9. Visualization of features extracted by well-trained DSC-KAN in 2 and 3 dimensions.

bus to obscure the distinction between attack and normal data, thereby resulting in a more overlapping data distribution between the two categories. These findings provide valuable insights into the DSC-KAN's capacity for extracting and encoding high-level, class-specific information. The well-defined clusters suggest that our proposed DSC-KAN has not only learned to classify samples accurately but has also developed a robust internal representation that captures the inherent structure of the CAN dataset.

Conclusion

This paper focuses on developing a data-driven IDS capable of learning the sequential patterns of in-vehicle network traffic and detecting both the presence and type of attacks based on traffic variations. We propose a deep learning-based IDS leveraging SN-GAN and DSC-KAN. To address the class imbalance between normal and abnormal data, we use SN-GAN for data augmentation. By generating diverse attack samples, SN-GANs can simulate various attack scenarios, thereby improving the model's generalization ability and robustness. Additionally, traditional CNNs are originally designed for large-scale image classification tasks, resulting in overly complex and large structures that are impractical for deployment in in-vehicle network anomaly detection. To overcome this, we integrate the latest advancements in deep learning by utilizing KAN convolutions, which offer powerful feature extraction capabilities, thus reducing the overall model size and complexity. Furthermore, we introduce DSC-KAN, a novel design inspired by depthwise separable convolutions, to further reduce the model's parameters. Our experiments involve four types of message injection attacks through the CAN bus, which are likely to occur in vehicles connected to external networks. We construct small sample datasets for each of these four publicly available attack types. The experimental results demonstrate that our proposed method accurately identifies the sequential patterns in in-vehicle network traffic data. Specifically, the IDS based on SN-GAN and DSC-KAN achieves reliable and efficient intrusion detection, with F1 scores of 0.99870, 0.99980, 0.99880, and 0.99990 for detecting fuzzy, gear spoofing, RPM spoofing, and DoS attacks, respectively. Finally, T-SNE visualizations reveal that the well-trained DSC-KAN not only classifies samples accurately but also develops a robust internal representation that captures the inherent structure of the CAN dataset.

Future research will focus on the following directions: (1) Given the high privacy and distributed nature of vehicle CAN bus data, we aim to explore the application of federal learning for vehicle network anomaly detection systems. This approach would enhance detection performance while safeguarding privacy by allowing collaborative learning without the need to centralize sensitive data. (2) We plan to investigate the integration of CAN bus data with other data sources, such as camera video streams, radar data, and GPS trajectories. By leveraging multimodal data in a collaborative learning framework, we aim to further improve the accuracy of anomaly detection systems.

Data availability

The datasets generated and analysed during the current study are not publicly available due to proprietary restrictions imposed by collaborating institutions but are available from the corresponding author on reasonable request.

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Conceptualization, Writing original draft, Review & Editing, W.Z. Software, Writing, Visualization, Y.Y. Data curation, Formal analysis, Experimentation, H.H. Supervision, Project administration, Y.C. Funding acquisition, Writing, F.Y.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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